Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Modeling Spring-Mass

1. Damped and Undamped Systems

Consider the horizontal spring-mass system. The motion of the block can be modeled using a linear 2nd-order ODE of the form

$$mx'' + bx' + kx = 0$$

where x is the block's displacement with constant parameters, mass $m > 0$, friction coefficient $b \geq 0$ and spring constant $k > 0$.

a. Determine the characteristic equation and its roots.

b. Determine the parameter conditions and the general solution of the homogeneous ODE for each case.

i. Overdamped (real distinct roots)

ii. Critically damped (real repeated roots)

iii. Underdamped (complex conjugate roots)

iv. Undamped (imaginary roots)

2. Systems with External Force

Consider the same horizontal spring-mass system shown in Problem (1). This time we add an external force acting on the mass.

a. Consider the undamped case ($b = 0$),

$$mx'' + kx = \cos(\omega t) + \sin(\omega t)$$

where $\omega \neq 0$ is a constant parameter. Use either undetermined coefficients or variation of parameters to solve for the particular solution. Then, determine the value of k for which the frequency of the homogeneous solution matches with the non-homogeneous term of the ODE.

b. Determine the general and specific solutions for Part (a) with initial conditions $x(0) = 1$, and $x'(0) = 0$.